# Mert Kilickaya

Senior AI Research Engineer

Amsterdam, Netherlands
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## Summary

Senior AI Research Engineer focused on foundation models and autonomous vision. Combines self-supervised learning, continual learning, and vision-language learning to develop agentic vision systems that improve through experience. Research aims to let models select useful data, derive supervision, and adapt to changes in the real world.

## Experience

**Senior Research Engineer** October 2026 – Present
*Greyparrot, London, UK (remote from the Netherlands)*

- Develop and evaluate visual foundation models for autonomous perception, with a focus on transferable representations.
- Research agentic vision systems that select useful observations and derive training supervision with reduced manual intervention.
- Investigate continual adaptation from new visual observations, with a focus on distribution shifts and catastrophic forgetting.

**Deep Learning Researcher** 2023 – October 2026
*Agendia, Netherlands*

- Led the design and development of a deep-learning-based biomarker for breast cancer recurrence, owning model architecture, validation strategy, and regulatory-facing documentation under IVDR constraints.
- Built and deployed end-to-end multiple-instance learning pipelines over tens of thousands of whole-slide images, covering data curation, training, evaluation, and containerized inference in PyTorch.
- Drove technical decision-making across the full ML lifecycle, collaborating with clinicians and biologists to align model behavior with clinical objectives and deployment requirements.
- Contributed to a filed patent, IVDR submission work, and research presented at ESMO 2026.

**Postdoctoral Researcher** 2022 – 2023
*TU Eindhoven, Netherlands*

- Conducted research on continual pre-training of foundation models under distribution shift, focusing on mitigating catastrophic forgetting in long-running self-supervised training regimes.
- Defined research directions and acted as a de facto co-supervisor for 5+ PhD and MSc students, leading to publications at ECCV and BMVC.

**Research Scientist Intern** 2020 – 2021
*Huawei, Finland*

- Improved large-scale mobile visual search systems under strict latency and memory constraints for on-device and near-device deployment.
- Developed text-interactive visual search models optimized for efficiency, resulting in an approved US patent.

**Graduate Research Assistant** 2017 – 2022
*Qualcomm QUVA Labs, Netherlands*

- Developed deep vision models for visual search, detection, and recognition.
- Contributed to research that resulted in 4 granted US patents and multiple top-tier publications.

## Education

**PhD in Deep Learning** 2017 – 2022
University of Amsterdam, Netherlands
Thesis: *Contextual Understanding of Visual Interactions*
Supervisor: *Prof. Arnold Smeulders*

**MSc in Computer Vision** 2013 – 2016
Hacettepe University, Turkey
Thesis: *Visual Importance with Applications to Vision and Language*
Supervisor: *Prof. Nazli Ikizler-Cinbis*

## Selected Publications

(h-index: 7, Citations: 370)
Top-tier venues including ECCV, BMVC, WACV, and EACL

- AI Driven Digital Pathology for the 70 Gene Recurrence Signature: Development and Validation in Early Stage Breast Cancer (ESMO 2026, Oral)
- HyTAS: A Transformer Architecture Search Benchmark (ECCV 2024)
- Locality-Aware ViTs for Hyperspectral Imaging (BMVC 2023)
- Are Labels Needed for Incremental Instance Learning? (CVPRW 2023, Oral)
- Structured Visual Search via Composition-Aware Learning (WACV 2021)
- HOI Detection via Weak Supervision (BMVC 2021)
- Re-evaluating Automatic Metrics for Image Captioning (EACL 2017, Oral)

## Patents

- Visual Search via Conversational Interaction (Huawei, 2022)
- Network for Interacted Object Localization (Qualcomm, 2021)
- Context-driven Learning of Human-Object Interactions (Qualcomm, 2020)
- Subject-Object Interaction Recognition Model (Qualcomm, 2019)

## Technical Skills

| Area | Expertise |
|---|---|
| **Research** | Visual foundation models, self-supervised learning, continual learning, vision-language learning |
| **Autonomous vision** | Data selection, supervision discovery, adaptation to distribution shifts, temporal learning |
| **Computer vision** | Object detection, classification, tracking, visual retrieval, whole-slide image analysis |
| **Model development** | PyTorch, transformer fine-tuning, multiple-instance learning, model evaluation |
| **Engineering** | Python, Docker, GPU training and inference, experiment tracking |

## Service & Awards

**Best Reviewer:** ECCV 2024

**Reviewer:** CVPR 2025, ECCV 2024, ICML, NeurIPS, ICLR 2023–2024, EMNLP 2021–2023